

Harshith Dupam

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// Technical Skills & Competencies

[Languages] Python, JavaScript (ES6+), C Language, GDScript (Godot Engine), SQL (PostgreSQL)

[Frontend UI] HTML5, CSS3, JavaScript Web APIs, Glassmorphic Frameworks, Event-Driven Architectures

[Systems Logic] Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), Finite State Machines

[Backend/Infra] Supabase Integration, Relational Database Schemas, Asynchronous Data Pipelines, File I/O

// Education History

Chaitanya Bharathi Institute of Technology (CBIT) 2025 - Present

Bachelor of Technology (B.Tech) in Chemical Engineering Core CGPA: 10.0 / 10.0

Core Academic Focus: Numerical Analysis, Programming, Engineering Mathematics, Data Structures

Narayana Junior College, Hyderabad, Telangana

Intermediate Higher Secondary Education MPC Framework (Mathematics, Physics, Chemistry)

// Technical Projects Portfolio

N//S//P — Neon Survival (Protocol React & Supabase)

- Architected an endless 2D arcade survival simulation application featuring complex rendering loops, engineered entirely over a responsive React core engine wrapper.
- Built a high-performance rendering canvas using HTML5 Canvas API to handle fluid 60FPS math-based coordinate tracking calculations for active entity instances.
- Integrated a secure, decoupled cloud database layer using Supabase client modules to handle asynchronous data synchronization for real-time user validation, player wallet tokens, and live global leaderboard schemas.
- Programmed a custom dynamic Finite State Machine (FSM) AI script that dictates enemy node behavior and sector boss tracking patterns.

DSA Visualizer Engine(Python / Tkinter)

- Developed an interactive data visualization platform using Python designed to step through execution stages of sorting (Bubble, Merge, Quick) and searching (Binary, Linear) algorithms.
- Resolved user interface freezing during high-iteration cycles by breaking recursive algorithm execution pipelines into distinct execution chunks, optimizing thread management.
- Constructed a robust diagnostic benchmark dashboard using Tkinter to evaluate, contrast, and visually graph Big-O complexity arrays.

Interactive Kanban Board Suite(JavaScript)

- Designed and implemented a functional task management dashboard leveraging native HTML5 Drag and Drop APIs to dynamically manage workflow cards across distinct operational stages.
- Configured persistent local cache tracking architectures to preserve column arrangement arrays and system data structures seamlessly across page refreshes.

Live Collaborative Whiteboard Architecture(JavaScript / Web APIs)

- Developed a shared design workspace engineered to capture continuous pointer vector input streams fluidly using HTML5 Canvas2D APIs.

Space Defenders Architecture(python / Pygame)

- Designed and compiled a complete space-shooter arcade title using the Pygame syntax.
- Implemented strict Object-Oriented Programming (OOP) structural design rules to completely decouple the underlying physics calculation engine loop from the interface rendering layers.
- Optimized memory footprints during intense gameplay states by engineering an entity pooling pipeline to systematically reuse and cycle laser/bullet instances.

Mini Coding Language Interpreter(Python)

- Architected a lightweight operational compilation pipeline consisting of a tokenizing lexer and sequential syntax parser built to interpret primitive variables and conditional loops.

Browser Storage Explorer(JavaScript)

- Built a local engineering audit deck designed to securely read, map, and inspect active memory registers, application caches, and session cookies directly within client runtimes.

Retro Snake Simulator(Python / Pygame)

- Programmed a standalone, customizable classic arcade engine using the Pygame framework running on strict coordinate matrix layouts.
- Developed custom self-referential grid checking logic to track structural intersections instantly, eliminating delay lag bugs.

Basic Calculator Binary Core(C Language)

- Compiled a highly optimal mathematical parser in native C language designed to handle complex calculation strings directly at the console level while enforcing explicit memory stack control constraints.

// Key Achievements & Contributions

- **Top-Tier Academic Standing:** Secured and maintained a perfect 10.0 CGPA framework throughout core engineering coursework at Chaitanya Bharathi Institute of Technology (CBIT).
- **Open Source Development:** Autonomously built, modularized, and actively managed a structural profile containing 12+ active technical code repositories on GitHub, showcasing strong version control habits.
- **Analytical Problem Solving:** Consistently applied algorithmic logic, computational methods, and systemic problem-solving frameworks across both software design challenges and chemical process modeling scenarios.